RICHARD ANTHONY "TONY" DEFAZIO, PhD

Slice Electrophysiology · Calcium Imaging · Analysis Infrastructure · AI-Assisted Research Software

tony@tonydefazio.com · github.com/syncytium2 · tonydefazio.com · Full CV: tonydefazio.com/cv.pdf

Cellular electrophysiologist and computational neurophysiologist with thirty years of brain-slice recording, from striatum and neocortex to hypothalamic GnRH and kisspeptin neurons, and building the rigs and analysis around it. First-author work covers excitability, voltage-gated potassium currents, GABAergic transmission, and dynamic clamp. Built and runs the Moenter lab's calcium-imaging analysis infrastructure on U-M Great Lakes and local workstations; since 2026, tonydefazio.com provides open-source AI-coding, education and data analysis tools, built with coding agents under an extensive review process.

EXPERIENCE

Associate Research Scientist, Molecular & Integrative Physiology — University of Michigan
Assistant 2012 → Associate 2018 → present · PI: Suzanne M. Moenter, PhD
Designed, built, and maintain the lab's imaging analysis infrastructure end to end: acquisition, U-M Great Lakes HPC batch processing, analysis, statistics, and the delivery layer. GnRH and kisspeptin neuron studies: patch clamp, voltage-gated current analysis, Kiss-GCaMP6f calcium imaging, dynamic-clamp conductance models.

Research Assistant Professor, Assistant Professor, Neurology — University of Miami · 2005–2011 · PI: Miguel Perez-Pinzon, PhD
Built a laser-speckle contrast imaging rig for real-time cerebral blood flow during laser-induced MCA branch occlusion in mice — optics, acquisition, and quantitative flow analysis. Electrophysiology and disease models: ischemic preconditioning, cardiac arrest, stroke, HIV-TAT synaptic effects.

Associate Scientist, Research Assistant Professor, Physiology — University of Miami · 2003–2005 · PI: Stephen D. Roper, PhD
Lingual-slice electrophysiology and calcium imaging of sour-taste sensory transduction.

Scientist — Johnson & Johnson Pharmaceuticals · 2002–2003
Preclinical screening of calcium-channel antagonists in an industry R&D setting.

Research Assistant Professor — University of Virginia · 2000–2002 · PI: Suzanne M. Moenter, PhD
Synaptic physiology, voltage-gated K^+ currents, excitatory GABA signaling in GnRH-GFP neurons.

Postdoctoral Research — UCLA (M. S. Levine) · UAB (J. J. Hablitz) · 1996–2000
Striatal and cortical physiology: patch clamp, calcium imaging, iontophoresis, voltage-sensitive dye imaging.

Across these roles: built and maintained integrated electrophysiology and calcium-imaging rigs end to end — hardware, acquisition, and analysis — and developed computer-aided analyses for current- and voltage-clamp data.

ELECTROPHYSIOLOGY: DATA AND SOFTWARE

Paired Electrophysiology and Imaging Ground Truth · 2024–2025

- Loose-patch action potentials with simultaneous calcium imaging of the same cell: 27 paired recordings, 5.7 hours, 17,947 action potentials, exported in the format of the published CASCADE spike-inference corpus for a MATLAB port of its network.
- In these recordings calcium events often occur without action potentials (unpublished), so we are testing if deep learning algorithms can be trained for spike inference under these conditions.

Patch-Clamp Analysis Suite in Igor Pro · 2018–2024 · 233 procedure files, ~84,000 lines (GitLab, private)

Hand-written over six plus years, before any coding agent existed, and in use across the lab for its patch-clamp and imaging data: action-potential properties, voltage-clamp event detection, sweep concatenation and batch processing panels, and the Igor Pro port of the CLUSTER pulse detector that nopeak.tonydefazio.com later reproduced in the browser.

gaba_excites — Current-Clamp Excitability Simulator in the Browser · React/Vite · gaba.tonydefazio.com · github.com/syncytium2/gaba_excites · MIT license

- Two published neuron models run client-side: a cortical pyramidal cell (Pospischil et al. 2008) ported from its ModelDB source, with a test that requires the same spike counts as NEURON running the original mechanisms and every spike within 0.25 ms; and a mouse GnRH neuron (Adams et al. 2018) transcribed from the paper, with a test that matches the published spike counts.
- Named for GABA's excitatory action in GnRH neurons, this was developed as an educational tool for students of the graduate Cell Physiology course at U-M.

Colonel Kernel — Calcium Convolution / Deconvolution in the Browser · Svelte/Vite · kernel.tonydefazio.com · github.com/syncytium2/colonel_kernel · MIT license

- Recovers calcium-transient kernels from user-uploaded traces with paired action potentials by regularized deconvolution; its spike-inference tab is deliberately limited to show where naive inference breaks down. Client-side only, so no data leaves the machine.
- Ported from the lab's MATLAB deconvolution code to Svelte/Vite through Claude Code with the full development history public, alongside a teaching tool with staged convolution and deconvolution exercises.

ANALYSIS INFRASTRUCTURE

Calcium-Imaging Data Pipeline — Moenter Lab, University of Michigan

- **Ingestion & motion correction:** series of up to ~44,000 frames in a proprietary format ingested via Bio-Formats; NoRMCorre motion correction with residual drift scored by a logged metric and a visual overlay.
- **Method comparison as infrastructure:** the lab's segmentation runs beside published algorithms EXTRACT, MIN1PIPE and CNMF-E on the same 134 recording windows, comparison in progress; six coordination detectors, three written in-house and three built on published methods.
- **Statistics & delivery:** multilevel data visualization (ROI, slice, group) with linear mixed-effects models in R, served by a plumber API to a Svelte/Vite front-end.

High-Throughput Batch Processing — U-M Great Lakes High Performance Computing

- **Resumable Slurm array:** one recording per task; the manifest is derived deterministically from the input folder and the worker skips finished files, so a failed task is rerun by resubmitting the same folder, not the array.
- **Memory sizing:** an analytic estimate first (~17.7 GB per single-precision copy of a 37k-frame, 387×310 stack, five alive at once), then a bytes-per-pixel-frame rule fitted to peak memory from Slurm accounting.
- **Pre-submit AI review:** a second Claude Code session, reviewing before submission, caught a path change that would have written multi-gigabyte outputs into the source repository.

OPEN-SOURCE PACKAGES

draughtsman — architecture figures for PyTorch models; `pip install draughtsman-nn`; DOI 10.5281/zenodo.22286341. `torch.jit.trace` supplies the operation graph, a coverage check fails if any traced operation is unaccounted for; nine models written out in full in-repo, including a hand-implemented Whisper-tiny.

bugarach — coordinated-activity detection ported from the lab's MATLAB, with a synthetic benchmark that plants events at known times so six detectors and small learned models are scored against what was there rather than against another detector's opinion; public raster viewer at bugarach.tonydefazio.com; BSD-3.

WORKING WITH CODING AGENTS

murderboard — Adversarial Document-Review Harness · Apache-2.0 license · DOI 10.5281/zenodo.22063835 · murderboard.tonydefazio.com

Runs eleven role-specialized AI reviewers over a document in parallel and merges their findings. A gate derives the roster from the document that defines the roles and fails any report missing a reviewer's findings, because a run of 7 roles and a run of 11 had produced reports no reader could tell apart.

short-course — Working with Coding Agents, for Scientists · github.com/syncytium2/short-course · lookedright.tonydefazio.com

A short course and the complete record of how it was made: the original eight points written before any AI was involved, the session that turned them into an outline, an eleven-role adversarial review (murderboard) of that outline (34 findings, 5 blocking), and what survived. Every claim the course made that turned out wrong after the murderboard is committed as a commit named after the defect. The repository's own public history is the friction log the course teaches.

TECHNICAL SKILLS

- **Neural signal acquisition:** whole-cell, perforated-patch, on-cell, outside-out and loose-patch recording; calcium and voltage-sensitive dye imaging (CCD, two-photon); laser-speckle cerebral blood flow; rig design, maintenance, and troubleshooting.
- **Signal processing & analytics:** spike-train and burst analysis, action-potential and synaptic-event detection, ion-channel kinetics, deconvolution and kernel estimation, coordinated-firing detection with synthetic ground truth.
- **Computation:** MATLAB (25+ years), Python/PyTorch, R, Igor Pro, JavaScript/Svelte; SLURM on U-M Great Lakes HPC.
- **AI-assisted development:** Claude Code daily; custom multi-agent review harnesses with mechanically verified role coverage.

PUBLICATIONS, TEACHING & EDUCATION

More than 60 peer-reviewed publications, 1996–2026, plus 3 book chapters. Full record: NCBI Bibliography. Selected, all first author:

- **Calcium imaging:** DeFazio, Burgess, Burger, Gibson & Moenter, *J Neuroendocrinol*, in press. Two types of calcium events in arcuate kisspeptin neurons.
- **Intrinsic excitability and dynamic clamp:** DeFazio, Navarro, Adams, Milescu & Moenter, *J Neurosci* 2019. Whole-cell potassium currents, dynamic-clamp GABA and AMPA conductances, and Markov models in arcuate kisspeptin neurons.
- **Synaptic transmission:** DeFazio, Elias & Moenter, *J Neurosci* 2014. GABAergic postsynaptic currents in kisspeptin neurons, with nonstationary fluctuation analysis.

Course Director, Cell Physiology (graduate), 2022–present: 18 lectures per academic year on quantitative physiology.

PhD, Biology — University of Southern California, 1996 · **BS, Physics** — Case Western Reserve University, 1989